

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

*I **Antony Magrin J(192572103)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Antony Magrin J

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CSA0713 – Computer Networks: Assessment Tool 2

Answers

Question 1: Smart Campus Subnetting (192.168.100.0/24)

Given Block: 192.168.100.0/24, allocated using VLSM in the order: Computing (/25) → Electronics (/26) → Mechanical (/27)

Subnet Allocation Table

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.127	192.168.100.1 – 192.168.100.126	$2^7-2 =$ 126
School of Electronics	/26	192.168.100.128	192.168.100.191	192.168.100.129 – 192.168.100.190	$2^6-2 =$ 62
School of Mechanical Engg.	/27	192.168.100.192	192.168.100.223	192.168.100.193 – 192.168.100.222	$2^5-2 =$ 30

Calculations (Working)

- **Computing (/25):** Host bits = $32-25 = 7 \rightarrow$ Block size = 128 $\rightarrow$ Range: .0 to .127
- **Electronics (/26):** Host bits = $32-26 = 6 \rightarrow$ Block size = 64 $\rightarrow$ Starts right after Computing's block at .128 $\rightarrow$ Range: .128 to .191
- **Mechanical (/27):** Host bits = $32-27 = 5 \rightarrow$ Block size = 32 $\rightarrow$ Starts at .192 $\rightarrow$ Range: .192 to .223

4. Remaining Unused IP Address Range

Total block = 192.168.100.0 – 192.168.100.255
Allocated so far = 192.168.100.0 – 192.168.100.223

Unused Range = 192.168.100.224 – 192.168.100.255 (a free /27 block of 32 addresses, available for future departments)

Question 2: Corporate Network Subnetting (10.10.0.0/16)

Given Block: 10.10.0.0/16, allocated using VLSM in the order: Software Dev (/24) → QA (/25) → HR (/26) → Executive (/27)

Subnet Allocation Table

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	$2^8 - 2 = 254$
Quality Assurance	/25	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	$2^7 - 2 = 126$
Human Resources	/26	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	$2^6 - 2 = 62$
Executive Management	/27	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	$2^5 - 2 = 30$

Calculations (Working)

- **Software Dev (/24):** Block size = 256 → occupies entire 10.10.0.0/24 → Range .0 to .255
- **QA (/25):** Next block starts at 10.10.1.0, block size = 128 → Range .0 to .127
- **HR (/26):** Next block starts at 10.10.1.128, block size = 64 → Range .128 to .191
- **Executive (/27):** Next block starts at 10.10.1.192, block size = 32 → Range .192 to .223

4. Remaining Unused IP Address Range

Total block = 10.10.0.0 – 10.10.255.255 (/16)

Allocated so far = 10.10.0.0 – 10.10.1.223

Unused Range = 10.10.1.224 – 10.10.255.255 (a very large pool remaining available for future departmental expansion)

Question 3: IPv4-to-IPv6 Migration for Smart Hospital

Given Address: 2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

- Remove leading zeros per group: 2001:db8:abcd:0:0:0:0:0
- Replace consecutive all-zero groups with ::

Compressed Address = 2001:db8:abcd::

2. Expanded (Full) Notation

Expanded Address = 2001:0db8:abcd:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits (first 4 hexets) define the network; remaining 64 bits are the Interface ID.

Network Prefix = 2001:0db8:abcd:0000::/64 (compressed: 2001:db8:abcd::/64)

4. & 5. Total Number of Host Addresses Supported

Host portion (Interface ID) = $128 - 64 = 64$ bits

Total Host Addresses = 2^{64} ($\approx 1.8 \times 10^{19}$ addresses)

(This astronomically large address space easily accommodates thousands of medical devices/IoT sensors with room for massive future growth — a key reason hospitals migrate to IPv6.)

Question 4: Routing Table Lookup

Routing Table:

Destination Network	Prefix
192.168.1.0	/24
192.168.2.0	/24
192.168.3.0	/24

Incoming Packet Destination IP: 192.168.2.45

1. Destination Subnet

Applying /24 mask (255.255.255.0) to 192.168.2.45 → Network portion = **192.168.2.0**

The packet belongs to subnet: 192.168.2.0/24

2. Matching Routing Table Entry

Matching Entry = 192.168.2.0/24 (Longest Prefix Match — exact match found in the table)

3. Next-Hop Network/Interface

The router forwards the packet out of the **interface directly connected to the 192.168.2.0/24 network** (i.e., the interface configured with an IP address in that subnet, e.g., 192.168.2.1) — since it is a directly connected route, no external next-hop router is needed.

4. Default Route (Fallback)

If the destination IP does **not** match any entry in the routing table, the router uses the **default**

route:

Default Route = 0.0.0.0/0 (Gateway of Last Resort) — this forwards the unmatched packet toward the ISP/upstream router for further routing.

Question 5: Layer-3 Router — Two Office LANs

Given:

- Administration LAN → 192.168.10.0/26 (Gateway: 192.168.10.1)
- Finance LAN → 192.168.10.64/26 (Gateway: 192.168.10.65)

1. Network Address & Broadcast Address

LAN	Subnet Mask	Network Address	Broadcast Address
Administration	/26 (255.255.255.192)	192.168.10.0	192.168.10.63
Finance	/26 (255.255.255.192)	192.168.10.64	192.168.10.127

2. Valid Host IP Address Range

LAN	Valid Host Range	Usable Hosts
Administration	192.168.10.1 – 192.168.10.62	$2^6 - 2 = 62$
Finance	192.168.10.65 – 192.168.10.126	$2^6 - 2 = 62$

3. Suggested IP Addresses for 3 Computers in Each LAN

LAN	PC1	PC2	PC3
Administration	192.168.10.2	192.168.10.3	192.168.10.4
Finance	192.168.10.66	192.168.10.67	192.168.10.68

(Note: .1 and .65 are reserved as the router gateway addresses for their respective LANs, so host assignment starts from .2 and .66 respectively.)

4. Default Gateway for Each LAN

LAN	Default Gateway (used by all hosts in that LAN)
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65

(Each host must be configured with its respective LAN's gateway IP so that the Layer-3 router can route inter-LAN traffic between Administration and Finance departments.)